

Problem Set: Regression

SOLUTIONS

1. (a) The **Multinomial Expansion** tells us how many terms we would have in a k -dimensional polynomial in p variables (see: https://en.wikipedia.org/wiki/Multinomial_theorem):

$$\binom{k+p-1}{p-1} = \frac{(k+p-1)!}{k!(p-1)!}$$

Here $k = 3$, $p = 6$, therefore :

$$\# \text{ terms} = \frac{(3+6-1)!}{3!5!} = 56$$

VERSION 1:

So our problem is **underdetermined**, and therefore not **unique**.

Explanation:

We have $m = 56$ parameters to learn, but only $n = 50$ data points (equations).

The OLS objective, L , is given by:

$$L = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2$$

Where $\mathbf{X}$ is the design matrix.

Note: $\mathbf{X} \in \mathbb{R}^{n \times m}$, $\mathbf{X}^T \mathbf{X} \in \mathbb{R}^{m \times m}$

Therefore, since $n < m$, $(\mathbf{X}^T \mathbf{X})$ is **rank deficient** and $(\mathbf{X}^T \mathbf{X}) \neq 0$

So L is not **strictly convex**.

By *Calculus Lecture: Quadratic Functions* this solution is not **unique** because L is not **strictly convex**.

VERSION 2:

So the normal equations will not yield a solution.

Explanation:

We have $m = 56$ parameters to learn, but only $n = 50$ data points (equations).

Recall from the **normal equations** that:

$$\mathbf{w}_{\text{OLS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Where $\mathbf{X}$ is the design matrix.

Note: $\mathbf{X} \in \mathbb{R}^{n \times m}$, $\mathbf{X}^T \mathbf{X} \in \mathbb{R}^{m \times m}$

Therefore, since $n < m$, $(\mathbf{X}^T \mathbf{X})$ is **rank deficient** and $(\mathbf{X}^T \mathbf{X})^{-1}$ does not exist.

Thus the normal equations do not yield a solution.

(b) Discard an attribute then $k = 3, p = 5$, therefore:

$$\# \text{ terms} = \frac{(3 + 5 - 1)!}{3!4!} = 35$$

Now we have more data points than unknowns.

$\mathbf{X}^T \mathbf{X}$ is invertible and the solution is unique and globally optimal, and the normal equations yield a solution.

(c) Regularisation

VERSION 1:

Explanation:

Adding an appropriate regulariser (e.g. the ℓ_2 -norm) ensures that the Hessian of the objective is positive definite.

Therefore the objective is strictly convex and our solution will be unique.

VERSION 2:

Explanation:

Via an Illustration, e.g. Ridge Regression (other illustrations acceptable):

Ridge Regression objective:

$$\tilde{L} = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

Therefore:

$$\mathcal{H} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}) \succ 0$$

Why? Because the rank of $(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})$ is always full due to the ridge.

(Relatedly, the revised normal equations:

$$\mathbf{w}_{\text{RR}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

yield a solution because $(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})$ is invertible.)

- (d) (No marks for any kind of suggestion which is equivalent to a combinatorial search!)

Use the LASSO:

$$\tilde{L} = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 + \frac{\lambda}{2} \|\mathbf{w}\|_1$$

Explanation: Second term in objective acts to push weights to zero (more aggressively than in ridge regression) as λ increases.

Why?

See lecture notes, especially demonstration that for subgradients around w_j then optimality obtains if:

$$-\frac{\lambda}{2} < \frac{\partial L}{\partial w_j} < \frac{\lambda}{2}$$

Thus the range of values of the gradient for which w_j remains at 0 if it is already there grows with λ .

2. (a)

$$\begin{aligned}
\mathbb{P}(\mathcal{S}; w) &= \prod_{i=1}^n \mathbb{P}(x^{(i)}, y^{(i)}; w) \quad \text{because of i.i.d. assumption} \\
&= \prod_{i=1}^n \mathbb{P}(y^{(i)} | x^{(i)}; w) \mathbb{P}(x^{(i)}) \quad \text{because } \mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B) \\
&= \prod_{i=1}^n \mathcal{N}(wx^{(i)}, \alpha) \mathbb{P}(x^{(i)}) \quad \text{by } y = wx + \varepsilon, \varepsilon \sim \mathcal{N}(0, \alpha) \\
&= \prod_{i=1}^n \mathbb{P}(x^{(i)}) \times \prod_{i=1}^n \frac{1}{\sqrt{2\pi\alpha}} \exp \left\{ -\frac{1}{2\alpha} (y^{(i)} - wx^{(i)})^2 \right\}
\end{aligned}$$

Therefore :

$$p_{\mathcal{X}, \mathcal{Y}}(\mathcal{S}; w) = \prod_{i=1}^n \frac{\mathbb{P}(x^{(i)})}{\sqrt{2\pi\alpha}} \exp \left\{ -\frac{1}{2\alpha} \sum_{i=1}^n (y^{(i)} - wx^{(i)})^2 \right\}$$

Therefore:

$$\begin{aligned}
A &= \prod_{i=1}^n \frac{\mathbb{P}(x^{(i)})}{\sqrt{2\pi\alpha}} \\
B &= -\frac{1}{2\alpha} \sum_{i=1}^n (y^{(i)} - wx^{(i)})^2
\end{aligned}$$

(b) Prior distribution:

$$p_{\mathcal{W}}(w) = \mathcal{N}(0, \beta) = \frac{1}{\sqrt{2\pi\beta}} \exp \left\{ -\frac{1}{2\beta} w^2 \right\}$$

Likelihood:

$$p_{\mathcal{X}, \mathcal{Y}}(\mathcal{S}|w) = p_{\mathcal{X}, \mathcal{Y}}(\mathcal{S}; w) = A \times \exp \left\{ -\frac{1}{2\alpha} \sum_{i=1}^n (y^{(i)} - wx^{(i)})^2 \right\}$$

Posterior distribution is $p_{\mathcal{W}}(w|\mathcal{S})$

By Bayes' Rule:

$$\begin{aligned} p_{\mathcal{W}}(w|\mathcal{S}) &\propto p_{\mathcal{X}, \mathcal{Y}}(\mathcal{S}|w)p_{\mathcal{W}}(w) \\ &= \frac{A}{\sqrt{2\pi\beta}} \exp \left\{ -\frac{1}{2\alpha} \sum_{i=1}^n (y^{(i)} - wx^{(i)})^2 - \frac{1}{2\beta} w^2 \right\} \\ &= \frac{A}{\sqrt{2\pi\beta}} \exp \left\{ -w^2 \left(\frac{1}{2\beta} + \frac{\sum_{i=1}^n (x^{(i)})^2}{2\alpha} \right) + w \left(\frac{\sum_{i=1}^n x^{(i)}y^{(i)}}{\alpha} \right) - \frac{\sum_{i=1}^n (y^{(i)})^2}{2\alpha} \right\} \\ &= \text{const.} \times \exp \left\{ - \left(\frac{\alpha + \beta \sum_{i=1}^n (x^{(i)})^2}{2\alpha\beta} \right) \left(w^2 - w \left(\frac{2\beta \sum_{i=1}^n x^{(i)}y^{(i)}}{\alpha + \beta \sum_{i=1}^n (x^{(i)})^2} \right) \right) \right\} \\ &= \text{const.}' \times \exp \left\{ - \left(\frac{\alpha + \beta \sum_{i=1}^n (x^{(i)})^2}{2\alpha\beta} \right) \left(w - \frac{\beta \sum_{i=1}^n x^{(i)}y^{(i)}}{\alpha + \beta \sum_{i=1}^n (x^{(i)})^2} \right)^2 \right\} \\ &= \text{const.}' \times \exp \left\{ -\frac{(w - \mu)^2}{2\sigma^2} \right\} \end{aligned}$$

where:

$$\begin{aligned} \mu &= \frac{\beta \sum_{i=1}^n x^{(i)}y^{(i)}}{\alpha + \beta \sum_{i=1}^n (x^{(i)})^2} \\ \sigma^2 &= \frac{\alpha\beta}{\alpha + \beta \sum_{i=1}^n (x^{(i)})^2} \end{aligned}$$

We recognise the characteristic form of a Gaussian distribution.

Since the posterior distribution is a true probability distribution then it must be normalised to 1.

This allows us to say that: $\text{const.}' = \frac{1}{\sqrt{2\pi\sigma^2}}$

More succinctly:

$$w|\mathcal{S} \sim \mathcal{N} \left(\frac{\beta \sum_{i=1}^n x^{(i)}y^{(i)}}{\alpha + \beta \sum_{i=1}^n (x^{(i)})^2}, \frac{2\alpha\beta}{\alpha + \beta \sum_{i=1}^n (x^{(i)})^2} \right)$$

- (c) The MAP estimate of w is the w for which the posterior, $p_{\mathcal{W}}(w|\mathcal{S})$ is maximal, i.e.:

$$w_{\text{MAP}} = \underset{w}{\operatorname{argmax}} \exp \left\{ -\frac{(w - \mu)^2}{2\sigma^2} \right\}$$

For a Gaussian distribution we know that the mode coincides with the mean, so:

$$w_{\text{MAP}} = \mu = \frac{\beta \sum_{i=1}^n x^{(i)} y^{(i)}}{\alpha + \beta \sum_{i=1}^n (x^{(i)})^2}$$

(d) Recall that our optimisation problem is:

$$\operatorname{argmax}_w \exp \left\{ -\frac{(w - \mu)^2}{2\sigma^2} \right\}$$

This can be written as:

$$\operatorname{argmin}_w \{(w - \mu)^2\}$$

Let us call the objective $L(w) = (w - \mu)^2$

Now:

$$\begin{aligned} \frac{\partial L}{\partial w} &= 2(w - \mu) \\ \frac{\partial^2 L}{\partial w^2} &= 2 > 0 \end{aligned}$$

By the definition of Strict Convexity (*Calculus Lecture: Convexity*) this satisfies the second order sufficient condition for strict convexity

Therefore L is strictly convex

By *Calculus Lecture: Convexity* then if L is strictly convex then the solution to: $\min_w L(w)$, must be unique

Therefore w_w is unique

(e) Recall from part (a) that:

$$p_{\mathcal{W}}(w|\mathcal{S}) = \text{const.} \times \exp \left\{ -\frac{1}{2\alpha} \sum_{i=1}^n (y^{(i)} - wx^{(i)})^2 - \frac{1}{2\beta} w^2 \right\}$$

From the definition of w_{MAP} :

$$\begin{aligned} w_{\text{MAP}} &= \underset{w}{\operatorname{argmax}} p_{\mathcal{W}}(w|\mathcal{S}) \\ &= \underset{w}{\operatorname{argmin}} \left\{ \frac{1}{2\alpha} \sum_{i=1}^n (y^{(i)} - wx^{(i)})^2 + \frac{1}{2\beta} w^2 \right\} \\ &= \underset{w}{\operatorname{argmin}} \left\{ \frac{1}{2} \sum_{i=1}^n (y^{(i)} - wx^{(i)})^2 + \frac{\alpha}{2\beta} w^2 \right\} \end{aligned}$$

This is identical to the canonical form of the optimisation problem associated with Ridge Regression:

$$\underset{w}{\operatorname{argmin}} \left\{ \frac{1}{2} \sum_{i=1}^n (y^{(i)} - wx^{(i)})^2 + \frac{\lambda}{2} w^2 \right\}$$

Where the connection is expressed via: $\lambda = \frac{\alpha}{\beta}$

3. (a) Ordinary Least Squares

(b)

$$\operatorname{argmin}_{\mathbf{w}} \frac{1}{2} \sum_{i=1}^n (y^{(i)} - \mathbf{w} \cdot \mathbf{x}^{(i)})^2 = \operatorname{argmin}_{\mathbf{w}} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2$$

(c) Note that:

$$\varepsilon \sim \mathcal{N}(0, \sigma^2) \quad \implies \quad y|\mathbf{x} \sim \mathcal{N}(\mathbf{w} \cdot \mathbf{x}, \sigma^2)$$

The likelihood is given by:

$$\begin{aligned} \mathbb{P}(\mathcal{S}; \mathbf{w}, \sigma) &= \prod_{i=1}^n p_{\mathcal{Y}}(y^{(i)}|\mathbf{x}^{(i)}; \mathbf{w}, \sigma) p_{\mathcal{X}}(\mathbf{x}^{(i)}) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y^{(i)} - \mathbf{w} \cdot \mathbf{x}^{(i)})^2}{2\sigma^2}\right) p_{\mathcal{X}}(\mathbf{x}^{(i)}) \end{aligned}$$

Let us search for the maximum likelihood estimator:

$$\begin{aligned} \mathbf{w}_{\text{MLE}} &= \underset{\mathbf{w}}{\operatorname{argmax}} \mathbb{P}(\mathcal{S}; \mathbf{w}, \sigma) \\ &= \underset{\mathbf{w}}{\operatorname{argmax}} \ln \mathbb{P}(\mathcal{S}; \mathbf{w}, \sigma) \\ &= \underset{\mathbf{w}}{\operatorname{argmax}} \ln \left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y^{(i)} - \mathbf{w} \cdot \mathbf{x}^{(i)})^2}{2\sigma^2}\right) \right) + \ln \left(\prod_{i=1}^n p_{\mathcal{X}}(\mathbf{x}^{(i)}) \right) \\ &= \underset{\mathbf{w}}{\operatorname{argmax}} \sum_{i=1}^n \left(-\frac{(y^{(i)} - \mathbf{w} \cdot \mathbf{x}^{(i)})^2}{2\sigma^2} \right) \\ &= \underset{\mathbf{w}}{\operatorname{argmin}} \sum_{i=1}^n (y^{(i)} - \mathbf{w} \cdot \mathbf{x}^{(i)})^2 \end{aligned}$$

(d) Note that:

$$\varepsilon \sim \text{Laplace}(0, b) \quad \implies \quad y|\mathbf{x} \sim \text{Laplace}(\mathbf{w} \cdot \mathbf{x}, b)$$

The likelihood is given by:

$$\begin{aligned} \mathbb{P}(\mathcal{S}; \mathbf{w}, \sigma) &= \prod_{i=1}^n p_{\mathcal{Y}}(y^{(i)}|\mathbf{x}^{(i)}; \mathbf{w}, \sigma) p_{\mathcal{X}}(\mathbf{x}^{(i)}) \\ &= \prod_{i=1}^n p_{\mathcal{Y}}(y^{(i)}|\mathbf{x}^{(i)}; \mathbf{w}, \sigma) p_{\mathcal{X}}(\mathbf{x}^{(i)}) \\ &= \prod_{i=1}^n \frac{1}{2b} \exp\left(-\frac{|y^{(i)} - \mathbf{w} \cdot \mathbf{x}^{(i)}|}{b}\right) p_{\mathcal{X}}(\mathbf{x}^{(i)}) \end{aligned}$$

Let us search for the maximum likelihood estimator:

$$\begin{aligned} \mathbf{w}_{\text{MLE}} &= \underset{\mathbf{w}}{\operatorname{argmax}} \mathbb{P}(\mathcal{S}; \mathbf{w}, \sigma) \\ &= \underset{\mathbf{w}}{\operatorname{argmax}} \ln \mathbb{P}(\mathcal{S}; \mathbf{w}, \sigma) \\ &= \underset{\mathbf{w}}{\operatorname{argmax}} \ln \left(\prod_{i=1}^n \frac{1}{2b} \exp\left(-\frac{|y^{(i)} - \mathbf{w} \cdot \mathbf{x}^{(i)}|}{b}\right) \right) + \ln \left(\prod_{i=1}^n p_{\mathcal{X}}(\mathbf{x}^{(i)}) \right) \\ &= \underset{\mathbf{w}}{\operatorname{argmax}} \sum_{i=1}^n \left(-\frac{|y^{(i)} - \mathbf{w} \cdot \mathbf{x}^{(i)}|}{b} \right) \\ &= \underset{\mathbf{w}}{\operatorname{argmin}} \sum_{i=1}^n |y^{(i)} - \mathbf{w} \cdot \mathbf{x}^{(i)}| \end{aligned}$$

- (e) The estimator is robust to outliers c.f. squared error

Because an outlying point is given less weight by the ℓ_1 norm than by the ℓ_2 norm

- (f) Note that the objective is not smooth and there is no analytic solution

Try numerical optimisation of some sort (proximal gradient descent, for example)

- (g) The objective function is convex, but it is not strictly convex (use the definition of convexity to demonstrate this).

Thus local minima are global minima, but such minima are not necessarily unique.